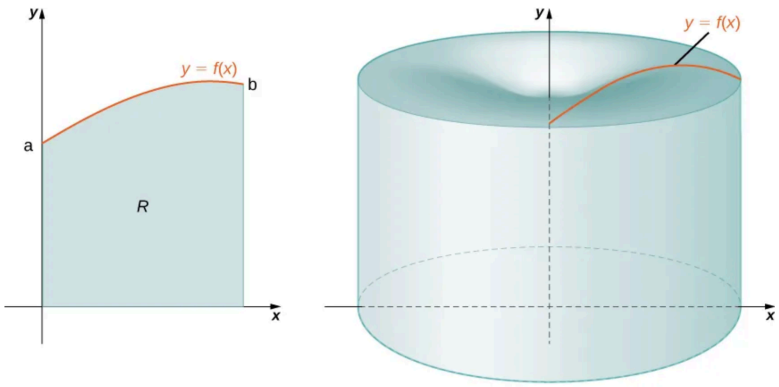
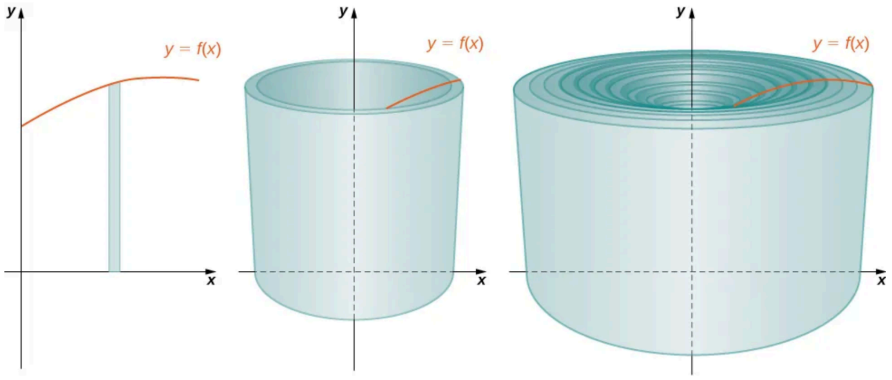
# 1 The Method of Cylindrical Shells

We now give a final method for finding the volume of a solid of revolution, called the *method of cylindrical shells*. With the disk and washer methods, one integrates along the coordinate axis parallel to the axis of revolution. With the method of cylindrical shells, one integrates along the coordinate axis *perpendicular* to the axis of revolution. This ability to choose which variable of integration to use can offer a significant advantage with more complicated functions.

Consider the plane region $R$ in the left figure below, which is revolved about the $y$-axis to form the solid of revolution to the right.




Choose a partition of the interval $[a,b]$ into $n$ subintervals of equal length $\Delta x$, and choose a sample point $x_i^* \in [x_{i-1}, x_i]$ in each subinterval. Construct a rectangle over the interval $[x_{i-1}, x_i]$ of height $f(x_i^*)$ and width $\Delta x$. When this rectangle is revolved about the $y$-axis, instead of a disk or washer we get a cylindrical shell. Putting all these shells together gives an approximation of the solid, as shown below:



Each shell is a cylinder, so its volume is the cross-sectional area multiplied by the height:

$$V_{\text{shell}} = f(x_i^*)\pi(x_i^2 - x_{i-1}^2). \tag{1}$$

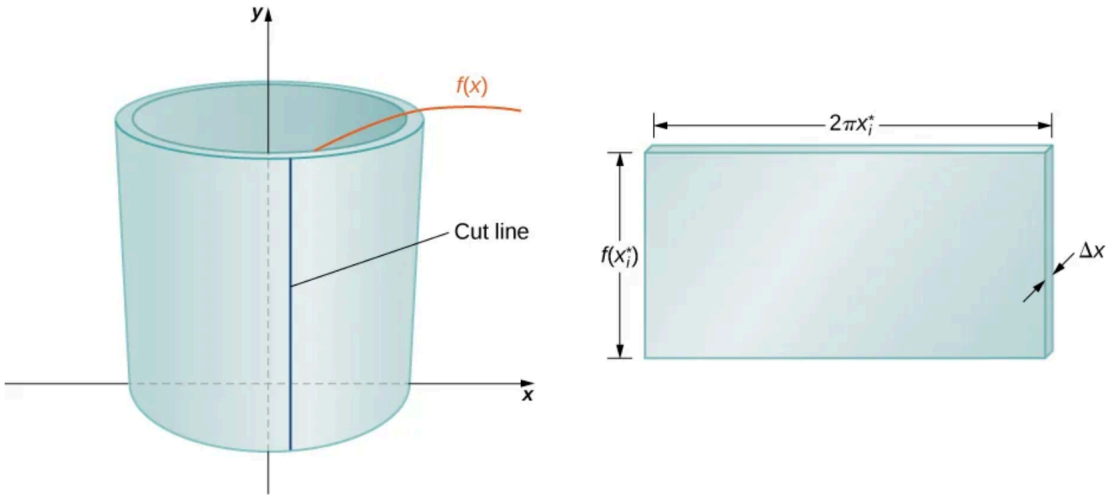
1. Show that $V_{\text{shell}}$ can be written as

$$V_{\text{shell}} = 2\pi f(x_i^*)\left(\frac{x_i + x_{i-1}}{2}\right)\Delta x. \tag{2}$$

Note that $\frac{x_i+x_{i-1}}{2}$ is the midpoint of the interval $[x_{i-1}, x_i]$ and the average radius of the shell, which we can approximate by $x_i^*$. We then have

$$V_{\text{shell}} \approx 2\pi f(x_i^*)x_i^*\Delta x.$$

Another way to think about this formula is as the volume of a thin plate obtained by making a vertical cut in the shell and unrolling it:



The volume of the solid is then approximately the sum of the volumes of the shells:

$$V \approx \sum_{i=1}^{n} 2\pi x_i^* f(x_i^*)\Delta x,$$

which is a Riemann sum for the function $2\pi x f(x)$. The exact volume is therefore

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi x_i^* f(x_i^*)\Delta x = \int_a^b 2\pi x f(x)dx.$$

We have proved the following:

**Theorem 1.3** (The method of cylindrical shells)**.** *Let $f(x)$ be continuous and nonnegative. Define $R$ as the region bounded above by the graph of $f(x)$, below by the x-axis, on the left by the line $x = a$, and on the right by the line $x = b$. Then the volume of the solid of revolution formed by revolving $R$ about the y-axis is given by*

$$V = \int_a^b 2\pi x f(x)dx.$$

2. Define $R$ as the region bounded above by the graph of $f(x) = \frac{1}{x}$ and below by the $x$-axis over the interval $[1,3]$. Find the volume of the solid of revolution formed by revolving $R$ about the $y$-axis.

3. Define $R$ as the region bounded above by the graph of $f(x) = 2x - x^2$ and below by the $x$-axis over the interval $[0,2]$. Find the volume of the solid of revolution formed by revolving $R$ about the $y$-axis.

As with the disk and washer methods, we can use the method of cylindrical shells with solids of revolution, revolved about the $x$-axis, when we want to integrate with respect to $y$.

4. Define $Q$ as the region bounded on the right by the graph of $g(y) = 2\sqrt{y}$ and on the left by the $y$-axis for $y \in [0,4]$. Find the volume of the solid of revolution formed by revolving $Q$ about the $x$-axis.

Suppose now we have a solid of revolution for which the graph of a function is revolved around a line other than one of the two coordinate axes. The formula

$$V_{\text{shell}} = 2\pi f(x_i^*)\left(\frac{x_i + x_{i-1}}{2}\right)(x_i - x_{i-1}) \tag{4}$$

was based on a solid revolved about the $y$-axis, which has an outer radius of $x_i$ and an inner radius of $x_{i-1}$. If we instead rotate the region around the line $x = -k$, then the outer radius will be $x_i + k$ and the inner radius $x_{i-1} + k$. Substituting these expressions into Equation (4) gives

$$\begin{aligned} V_{\text{Shell}} &= 2\pi f(x_i^*)\left(\frac{(x_i + k) + (x_{i-1} + k)}{2}\right)((x_i + k) - (x_{i-1} + k)) \\ &= 2\pi f(x_i^*)\left(\frac{x_i + x_{i-1} + 2k}{2}\right)(x_i - x_{i-1}) \\ &= 2\pi f(x_i^*)\left(\left(\frac{x_i + x_{i-1}}{2}\right) + k\right)\Delta x. \end{aligned}$$

As before, since $\frac{x_i + x_{i-1}}{2}$ is the midpoint of the interval $[x_{i-1}, x_i]$, it can be approximated by $x_i^*$, and therefore

$$V_{\text{Shell}} \approx 2\pi(x_i^* + k) f(x_i^*)\Delta x.$$

It follows that

$$\begin{aligned} V &= \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi(x_i^* + k) f(x_i^*)\Delta x \\ &= \boxed{\int_a^b 2\pi(x + k) f(x)\,dx}. \end{aligned}$$

5. Define $R$ as the region bounded above by the graph of $f(x) = x$ and below by the $x$-axis over the interval $[1,2]$. Find the volume of the solid of revolution formed by revolving $R$ about the line $x = -1$.

We could also rotate the region about other horizontal or vertical lines. In each case, the volume formula must be adjusted accordingly.

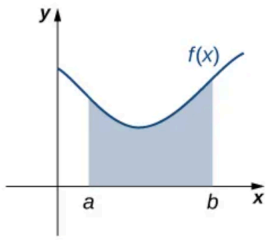
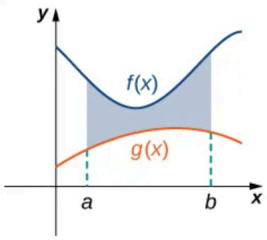
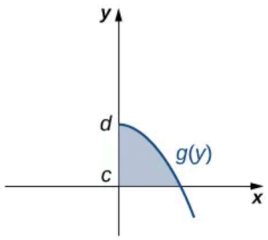
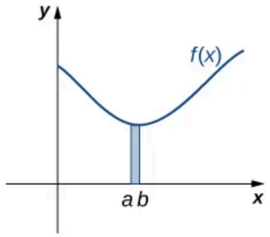
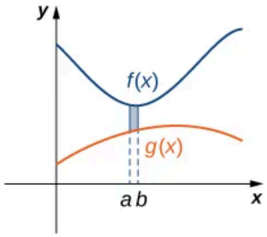
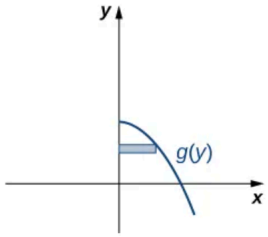
For our final example, let's look at the volume of a solid of revolution for which the region of revolution is bounded by the graphs of two functions.

6. Define $R$ as the region bounded above by the graph of the function $f(x) = \sqrt{x}$ and below by the graph of the function $g(x) = \frac{1}{x}$ over the interval $[1,4]$. Find the volume of the solid of revolution generated by revolving $R$ about the $y$-axis.

# 2 Choosing which method to use

We have studied several methods for finding the volume of a solid of revolution, but how do we know which method to use? If often comes down to a choice of which integral is easiest to evaluate. The figure below describes the different approaches for solids of revolution about the $x$-axis. There is an analogous table for solids of revolution about the $y$-axis.

Comparing the Methods for Finding the Volume of a Solid Revolution around the $x$-axis

| Compare | Disk Method | Washer Method | Shell Method |
|---|---|---|---|
| Volume formula | $V = \int_a^b \pi[f(x)]^2\,dx$ | $V = \int_a^b \pi[(f(x))^2 - (g(x))^2]\,dx$ | $V = \int_c^d 2\pi y\, g(y)\,dy$ |
| Solid | No cavity in the center | Cavity in the center | With or without a cavity in the center |
| Interval to partition | $[a, b]$ on $x$-axis | $[a, b]$ on $x$-axis | $[c, d]$ on $y$-axis |
| Rectangle | Vertical | Vertical | Horizontal |
| Typical region |  |  |  |
| Typical element |  |  |  |

7. For each of the following problems, select the best method to find the volume of a solid of revolution generated by revolving the given region about the $x$-axis, and set up the integral to find the volume (but do not evaluate the integral).

   (a) The region bounded by the graphs of $y-x, y=2-x$, and the $x$-axis.

   (b) The region bounded by the graphs of $y=4x-x^2$ and the $x$-axis.